

e2m2e White Paper

An Algorithm Toolset for Cislunar Mission Planning:
Architecture, Algorithms, and Verification

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<https://github.com/cislunarspace/e2m2e>

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Abstract

e2m2e (Earth to Moon, Moon to Earth) is an open-source algorithm toolset for cislunar mission planning. In an LLM+Agent-style autonomous mission planning system, the large language model understands mission intent and decomposes/orchestrates subtasks, while e2m2e provides precise and reliable orbit computation: it builds dynamical models of cislunar space, generates periodic orbit families, designs transfer paths between orbits, and visualizes results for inspection. This white paper presents the system architecture (a layered Python-orchestration + Rust-numerics design), the core algorithms (differential correction, multiple shooting, continuation, invariant-manifold stitching, Lambert/low-energy/low-thrust transfers), the spacetime reference and verification framework, and the three calling interfaces (Facade, CLI, and MCP).

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1 Introduction

1.1 The Computational Demands of Cislunar Mission Planning

With the Artemis program, the Gateway station, the Chang’e lunar exploration program, and commercial lunar missions advancing, cislunar space has become a focal region for mission design. Unlike near-Earth orbits, cislunar dynamics are dominated by the combined gravity of the Earth and the Moon: periodic orbits near the libration points (Halo, NRHO, DRO, and others) have no analytic solution, and their design relies on numerical methods in the three-body framework—differential correction, multiple shooting, parameter continuation, and invariant manifold theory—while transfer design between orbits must trade off impulsive, low-energy, and low-thrust propulsion modes. Such computations share three traits: **many models** (the circular restricted three-body problem CR3BP, ephemeris N -body, and the bicircular restricted four-body problem BCR4BP coexist at different fidelities), **heavy iteration** (a single ephemeris correction often involves tens of thousands of trajectory segments), and **long chains** (initial guess, correction, high-fidelity propagation, and control simulation are tightly coupled).

1.2 Positioning: A Numerical Foundation for the LLM+Agent Era

e2m2e (Earth to Moon, Moon to Earth) addresses these needs as an **algorithm toolset infrastructure for cislunar mission planning**. In an LLM+Agent-style autonomous mission planning system, the large language model understands mission intent and decomposes/orchestrates subtasks; e2m2e handles the numerical half—building dynamical models of cislunar space, generating periodic orbit families, designing transfer paths between orbits, and visualizing results for inspection. Beyond the conventional Python API and CLI, e2m2e natively provides an MCP (Model Context Protocol) server: 13 task-level tools are exposed to LLM agents over stdio, the tool list is derived from Facade method metadata, artifacts are automatically archived, and `record_ids` chain across tools—so that “design an NRHO and run 100 Monte Carlo station-keeping simulations on it” becomes a single executable instruction in natural language.

1.3 Relationship to Existing Tools

- **GMAT / STK**: general-purpose mission analysis tools, driven mainly by GUIs and scripts, with thorough validation data but hardly callable programmatically by agents;
- **Orekit / poliastro**: general-purpose astrodynamics libraries covering near-Earth to deep space, but not centered on the cislunar three-body problem and offering no task-level encapsulation;
- **e2m2e**: focused on cislunar space, encapsulating the entire design chain—family selection, initial guess, correction, high-fidelity propagation, transfer, station keeping—as task-level interfaces, with a Rust numerical kernel and an agent-native MCP interface.

e2m2e does not aim to replace general-purpose tools but complements them: propagation accuracy is aligned with GMAT and DFH to the sub-100 m level (Section 9), the coordinate and time chains are cross-validated against GMAT reference data, and it offers what general tools lack—periodic orbit family generation and an agent-native interface. Section 8 develops a systematic comparison against ten peer and ecosystem tools based on a code-level survey.

1.4 Organization of This Report

Section 2 presents the system architecture: five architectural modules and a layered “Python orchestration + Rust numerical kernel” runtime. Section 3 covers the spacetime reference, dynamical models, and high-fidelity force models. Sections 4 and 5 present the core algorithms for

periodic orbit design and transfer design. Section 6 covers orbit control and Monte Carlo simulation. Section 7 describes the Facade/CLI/MCP interfaces and the orbit catalog. Section 8 positions e2m2e against ten tools including GMAT, STK, ATK, Orekit, and pykep. Section 9 presents the verification framework and concludes with an outlook.

2 System Architecture

2.1 Starting from One NRHO Orbit

Architecture decisions are rarely made in the abstract. Take “design an Earth–Moon L2 NRHO” as an example: a single mission request passes through every part of the system and exposes exactly why each layer exists:

- **Time.** The user enters 2024-01-01 00:00:00 UTC, but the independent variable of DE ephemerides is barycentric dynamical time TDB, about 69 seconds ahead of UTC at that moment. The Moon orbits Earth at roughly 1 km/s, so querying its position with UTC taken as TDB misplaces it by about 70 km—the lunar braking window is lost entirely. e2m2e therefore standardizes on TDB internally and converts to UTC only at interface boundaries, the same practice as STK.
- **Frames.** The CR3BP initial guess is given in the Earth–Moon synodic frame, mission specifications live in the J2000 inertial frame, and ground stations in the ITRF93 Earth-fixed frame. The same (x, y, z) numbers in another frame mean another physical location, and the compiler will not flag such errors (the 1999 Mars Climate Orbiter unit confusion is the same class of lesson).
- **Constants.** Earth’s GM has DE440, DE421, and WGS-84 values; so does the Earth–Moon mass ratio μ . e2m2e itself stumbled here: Python and Rust each copied a μ , and the computed orbit families mismatched the literature. Constant baselines (multiple coexisting sets, self-consistent within each, single-sourced across both languages) exist for exactly this.
- **Scale.** A 2-year 50-rev ephemeris correction involves roughly twenty thousand trajectory segments, thousands of steps per segment, force-model evaluation and ephemeris queries at every step—billions of scalar operations. This layer must be fast and parallelizable; since CSPICE kernels are global state and cannot be used concurrently, the Rust layer holds no SPICE handles and consumes only pre-sampled injected ephemeris cache tables.
- **Orchestration.** DRO or Halo, how large an initial-guess amplitude, how tight a tolerance, how long a propagation—these are domain decisions that change daily, and they stay in Python.

2.2 The Runtime Layers

The runtime architecture is just four pieces, plus two supporting ones (Figure 1):

- **e2m2e/api:** the sole external entry. A task-level Facade from which the CLI and the MCP server are derived from the same metadata source;
- **e2m2e/algorithm:** constructs problems with domain knowledge—choosing orbit families, constraints, and initial guesses (seeds from catalog records or the normal-form module). It does only three things: construct the problem, call the Rust iterators, and interpret results—no numerical iteration;
- **crates/:** the Rust numerical layer, where heavy iterations converge. Six crates: spice (CSPICE FFI + caching), propagation (pure-math integrators), forces (force models + STM), integrators (pyo3 bindings + iterative solvers such as shooting + parallelism),

levelset (level-set/HJB solver kernel), and hjb-dynamics (Hamiltonian dynamics for the HJB solver);

- **e2m2e/data**: ephemeris cache tables, frame data (EOP/leap seconds), and constant baselines;
- **e2m2e/tools**: logging and visualization support; **e2m2e/mbse**: MBSE data models and diagram generation, outside the dependency chain.

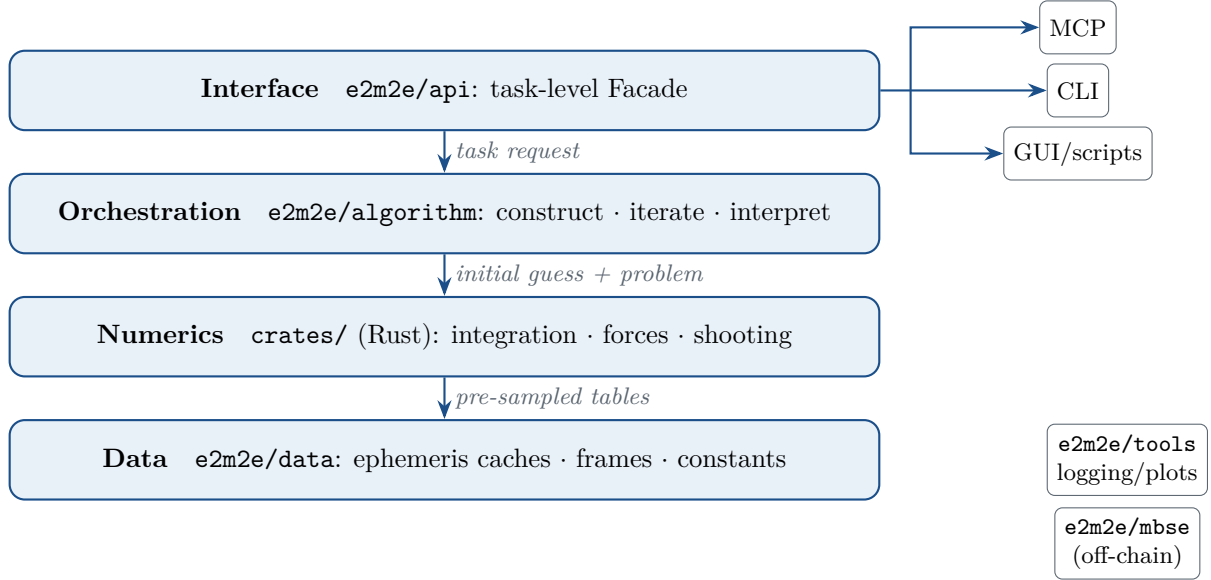


Figure 1: Runtime layers of e2m2e. Dependencies point downward only: the CLI and MCP derive from the same Facade source; the orchestration layer constructs problems but never iterates; the Rust numerical layer holds no SPICE handles and receives ephemerides as pre-sampled cache tables.

The journey of one orbit task: **api** receives the request → **algorithm/family** picks a family and initial guess → shooting sinks into the **crates** integrators, with per-step forces evaluated by the forces crate (ephemerides from the pre-sampled cache tables of **data/frames**, constants from **data/constants**) → the result lands in the **catalog/** orbit catalog and is delivered through the CLI/MCP.

2.3 Five Architectural Modules

Above the runtime layers, the engineering organization is divided into five architectural modules (Table 1). The two are different viewpoints: the layers describe how computation flows; the modules describe how the project is maintained.

Design Note

The boundary of numerical sinking is decided by iteration density: shooting, continuation, NSGA-II evolutionary operators, and low-thrust direct-method kernels already live on the Rust side; Python keeps outer orchestration and reference paths. The constraint that the Rust layer holds no SPICE handles makes the whole numerical kernel inherently thread-safe and parallelizable with Rayon. Per-module migration progress is itemized in [docs/architecture/numerics-migration-status.md](#).

3 Spacetime Reference and Dynamical Kernel

Table 1: The five architectural modules

Module	Responsibilities
Spacetime & constants	Time scales (UTC/TDB/TAI/TT; dynamics standardized on TDB); frame conversion (J2000/ITRF93/IAU 2006, GCRS–EBCRS, dynamic axes); physical constant baselines (DE421/DE440/WGS84 coexisting, single-sourced across Python/Rust against drift)
Rust computation	Six crates (above); iterative problems are constructed by Python and passed in, Rust only iterates to convergence; ephemerides injected as cache tables, enabling parallelism
Python orchestration	Construct problems, call Rust iterators, interpret results; three tiers: task-level Facade, design chains (family initial guess → Rust correction → high-fidelity propagation), subproblem construction
CI	Three-platform wheel matrix (Windows x64, Linux x86_64, Linux aarch64) + sdist + GitHub Release + PyPI; CSPICE build packages ship via GitHub Release; the manylinux 2_28 glibc baseline covers domestic distributions
Data management	GitHub Release manages build libraries and ephemeris data; Git tracks orbit-family seed parameters; local artifacts enter the orbit catalog (JSON metadata + NPZ array segments, SQLite only as a derived index)

3.1 Spacetime Reference: Time, Frames, Constants

Orbit-computation errors often originate not in algorithms but in reference baselines. e2m2e maintains the spacetime reference as a dedicated module:

Time. Internally, dynamics standardize on barycentric dynamical time TDB (SPICE ET seconds or JD_TDB); UTC appears only at interface boundaries. The conversion chain covers UTC / TAI / TT / TDB, with leap-second kernels loaded automatically.

Frames. A coordinate system is decomposed into **Axes + Origin**: given an epoch, the axes return a rotation matrix and the origin returns its absolute state; together they form a complete mathematical reference frame. Axes types cover the ICRS inertial frame, SPICE high-precision ITRF93 (verified against `pxform/sxform` to 10^{-12}), a GMAT-compatible ITRF, IAU 2006 simplified Earth orientation (for SPICE-free scenarios), and the dynamic axes VNB and LVLH that depend on the spacecraft’s instantaneous state (customary for thrust directions and station keeping). Batch conversion between the synodic frame and J2000 has sunk to Rust, including the barycenter→geocenter translation correction.

Joint spacetime transformation. Built on the `r2s2` library (a cislunar spacetime coordinate library), the TDT+GCRS ↔ TDB+EBCRS transformation includes the relativistic terms required by IAU resolutions; e2m2e fills the EBCRS origin gap ($\mathbf{x}_{\text{ebcrs}} = \mathbf{x}_s - \mathbf{x}_{\text{emb}}(t)$). Round-trip verification: position difference < 0.001 mm, time difference < 1 μs ; relativistic corrections amount to about 0.20 m at LEO altitude and 10.1 m at lunar distance.

Constants. Physical constants are managed as baselines (DE421 / DE440 / WGS-84 coexisting, each internally self-consistent, chosen per scenario), generated single-sourced for both Python and Rust to prevent drift. The Earth–Moon CR3BP baseline takes the DE421 self-consistent values: $\mu = 0.01215058535$, characteristic length DU = 384400 km, characteristic time TU ≈ 375190 s (≈ 4.34 days).

3.2 Three Layers of Dynamical Models

e2m2e provides three dynamical models at increasing fidelity, with state conversion among them (Table 2).

The circular restricted three-body problem (CR3BP) is a dimensionless autonomous system

Table 2: Three layers of dynamical models

Model	Fidelity	Use and properties
CR3BP	Concept design	Dimensionless autonomous system in the synodic frame, with the Jacobi integral and five libration points; the carrier of periodic orbit family design
BCR4BP	Intermediate	Adds an analytic solar perturbation (Sun position as an analytic function of time, no ephemeris needed); time-periodic; the carrier of WSB searches
Ephemeris N -body	High fidelity	J2000 inertial frame in physical units, N configurable bodies queried from SPICE ephemerides; accurate mission-level extrapolation

in the synodic rotating frame:

$$\ddot{x} - 2\dot{y} = \frac{\partial \Omega}{\partial x}, \quad \ddot{y} + 2\dot{x} = \frac{\partial \Omega}{\partial y}, \quad \ddot{z} = \frac{\partial \Omega}{\partial z}, \quad (1)$$

with pseudo-potential and gravitational distances

$$\Omega = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2}, \quad r_1 = \sqrt{(x+\mu)^2 + y^2 + z^2}, \quad r_2 = \sqrt{(x-1+\mu)^2 + y^2 + z^2}, \quad (2)$$

and the only integral of motion, the Jacobi constant $C_J = 2\Omega - v^2$.

The bicircular restricted four-body problem (BCR4BP) superposes a solar point-mass perturbation (direct + indirect terms) on the CR3BP, with the Sun position given analytically, $\mathbf{r}_s(t) = a_s(\cos \theta, \sin \theta, 0)$, $\theta = \theta_0 + \omega_s t$, retrograde in the synodic frame. The system is no longer autonomous but remains time-periodic, and serves as the working medium for WSB ballistic-capture searches: against the ephemeris model, the 1-day extrapolation deviation is about 10^3 km, and it outperforms the CR3BP within 2 days.

The ephemeris N -body model takes a central point mass plus third-body perturbations:

$$\mathbf{a} = -\frac{\mu_0 \mathbf{r}}{\|\mathbf{r}\|^3} - \sum_i \mu_i \left[\frac{\mathbf{r} - \mathbf{r}_i}{\|\mathbf{r} - \mathbf{r}_i\|^3} + \frac{\mathbf{r}_i}{\|\mathbf{r}_i\|^3} \right], \quad (3)$$

with body states from SPICE ephemerides—non-autonomous, without libration points or a Jacobi integral—for mission-level precise design. All three layers support 42-dimensional augmented STM propagation ($\delta \mathbf{x}(t) = \Phi(t, t_0) \delta \mathbf{x}(t_0)$).

3.3 High-Fidelity Force Models

On top of the ephemeris layer, force models are managed as a composable container with registration and per-entry enable/disable; each component provides an analytic Jacobian and a Rust serialization hook:

- **Gravity:** central point mass; third-body direct + indirect terms; fully normalized spherical-harmonics gravity field (Pines recursion, supporting EGM96 / GRGM900C and custom coefficient files, with built-in Earth solid tides and a lunar $k_2 = 0.024116$ solid tide);
- **Solar radiation pressure:** cannonball model ($P_{1\text{AU}} = 4.56 \times 10^{-6}$ N/m², optional conical shadow model with multi-occulter composition aligned with GMAT) and the ECOM 9-parameter empirical model (DFH-compatible);
- **Atmospheric drag:** computed in ITRF with default $C_d = 2.2$; density from the USSA76 piecewise-exponential atmosphere with first-order F10.7 / Ap corrections;

- **Thrust:** continuous thrust (direction frame selectable among inertial / VNB / LVLH) and variable-mass continuous thrust (mass as a 7th state component, 7-dimensional augmented Rust path); impulsive maneuvers applied piecewise;
- **Relativistic correction:** post-Newtonian Schwarzschild + Lense–Thirring + de Sitter terms, aligned with GMAT.

Accuracy: full force-model propagation aligns with GMAT and DFH to the sub-100 m level; against a full DFH configuration, a 7-day Halo arc deviates by 88 m and a 7-day Lissajous arc by 201 m, with 97% of the residual due to the intrinsic difference between the ECOM and cannonball SRP models. Per ADR 0013, GMAT/DFH serve only as development-time cross-references; correctness is arbitrated by physical definition.

3.4 The Rust Integrator Kernel

All heavy iteration runs on the Rust side, which provides three integrator families (Table 3). Adaptive RK error control uses the threshold $\text{tol} \cdot \max(1, \|y\|)$; augmented STM components do not participate in step-size control (consistent with GMAT). Benchmark (normalized LEO+J2, 1 day, $\text{tol} = 10^{-13}$): PD45 takes about $5000 \text{ steps} \times 7 \text{ evaluations}$, RK89 about $800 \text{ steps} \times 16 \text{ evaluations}$, and ABM about $500 \text{ steps} \times 2 \text{ evaluations}$ —higher-order and multistep methods significantly reduce the total number of right-hand-side evaluations.

Table 3: The three Rust integrator families

Family	Methods	Order	Typical use
Adaptive single-step RK	PD45 / PD78 / RK89	5 / 8 / 9	General propagation and accuracy baselines
Adams multistep	ABM (PECE)	4	Long arcs with smooth dynamics; only 2 RHS evaluations per step
Cowell double integration	Störmer–Cowell	8	Pure-gravity orbits; state dimension halved

Two performance-critical designs: first, the **ephemeris cache**—SPICE queries are pre-sampled on a uniform grid and injected into Rust as cubic splines (C^2 continuous), accelerating full force-model propagation by more than $10\times$ while removing the CSPICE global-state constraint on parallelism; second, the **compiled path**—once the force model is serialized, the entire integration loop runs inside Rust with no Python round-trips. Event detection follows scipy semantics (**terminal** / **direction**), and Poincaré sections can directly generate event functions for in-integration detection.

4 Periodic Orbit Design

4.1 An Overview of Orbit Families

Usable cislunar orbits fall into four layers by distance from the Moon, with scale as the main thread: the farther from the Moon, the larger Earth’s share of the gravity field, the dynamics transition from two-body Keplerian to three-body, and orbits turn from stable to unstable (Table 4). e2m2e provides a unified family-generation entry, `orbit_family_generation()`, covering eight orbit families; Lyapunov and resonant orbits (RO) are additionally supported at the differential correction strategy layer.

Periodicity semantics are strictly distinguished: Halo, NRHO, Axial, SPO, LPO, Horseshoe, and DRO members are strictly periodic; Lissajous members are quasi-periodic bounded trajectories and are not forced into periodic closure.

Table 4: Periodic orbit families

Family	Location	Characteristics and uses
DRO	Moon-centered	Distant retrograde orbit; neutrally stable with low maintenance cost; candidate for lunar orbital stations
DPO	near L2	Distant prograde orbit; unstable; usable for far-side communication relay
Halo / NRHO	L1/L2	Three-dimensional periodic orbits; NRHOs are the high-amplitude near-rectilinear form ($A_z > 0.1$ dimensionless), adopted by Gateway
Axial	L1/L2	Gómez Type B bifurcation; axial motion, coverage complementary to Halo
Lyapunov	L1/L2/L3	Planar periodic orbits
Lissajous	L1/L2/L3	Quasi-periodic (incommensurate in-plane/out-of-plane frequencies), generated on the nonlinear center-reduced flow
SPO / LPO	L4/L5	Short-/long-period orbits; large-amplitude LPOs take a horseshoe shape
Horseshoe	L4/L5	Large-amplitude members of the LPO chain
RO	—	Resonant orbits, departing from the y -axis symmetry

4.2 Initial Guesses: Starting from Analytic Approximations

The first building block of periodic orbit design is the initial guess, for which e2m2e has three sources:

- **Richardson third-order analytic approximation** (1980): a third-order analytic solution for Halo orbits near the collinear libration points. The internal pipeline: solve the libration-point distance parameter γ (quintic equation, Brent’s method) $\rightarrow$ compute the in-plane frequency $\omega_p \rightarrow$ assemble the third-order coefficients and solution. Small-amplitude seeds (e.g. dimensionless $z_0 = 0.001$) are the most accurate and are then amplified by continuation;
- **Seed orbits**: the DRO family starts from a standard seed (amplitude ≈ 90786 km), differentially corrected with the x -axis crossing fixed and continued bidirectionally, covering an envelope of 1737–110000 km;
- **Normal-form characteristic parameters**: the dynamics near a libration point are reduced to action–angle characteristic parameters $[q_1, p_1, I_2, \theta_2, I_3, \theta_3]$, describing with a few constants an orbit that would otherwise need six state components evolving in time. The pipeline: ephemeris initial state $\rightarrow$ dynamical substitute orbit $\rightarrow$ quasi-Floquet transformation $\rightarrow$ center-manifold reduction $\rightarrow$ characteristic parameters. The quasi-Floquet reduction offers a 36-dimensional matrix method or a 21-dimensional $\mathfrak{sp}(6)$ Lie-algebra parameterization (symplectic by construction); the center-manifold Lie transformation is truncated at order 10 by default.

4.3 Differential Correction: STM Newton Iteration via Symmetries

Differential correction iteratively adjusts the initial conditions so that the orbit satisfies periodicity constraints, at its core an STM-based Newton iteration. e2m2e exploits the three mirror symmetries of the CR3BP to shrink the problem—for example, XZ-plane symmetry means that if $(x, y, z, \dot{x}, \dot{y}, \dot{z})$ is a solution, so is $(x, -y, z, -\dot{x}, \dot{y}, -\dot{z})$; a Halo orbit therefore only needs to depart perpendicularly from the XZ plane ($y = 0$) and be corrected to cross it perpendicularly again half a period later.

Different combinations of symmetry and fixed parameters form 11 correction strategies (Lyapunov/DRO with fixed x_0 , fixed-period orbits with fixed $T_{1/2}$, RO with fixed y_0 , Halo

with fixed z_0 or x_0 , Axial with fixed v_{z0} , SPO/LPO at L4/L5 without symmetry and full-period closure, etc.). Architecturally, the strategy layer is strictly separated from the solver: strategy functions only produce immutable configuration objects, and execution sinks to the Rust CR3BP correction kernel.

Convergence failures are diagnosable: the history records per-step errors and the termination cause (converged / correction below machine precision / diverged / singular Jacobian / stagnated / converged to a parasitic root with $T \approx 0$); a singular Jacobian indicates linearly dependent constraints and calls for a different strategy.

4.4 Multiple Shooting: Ephemeris Correction of Long Arcs

Differential correction is sensitive to the initial guess, increasingly so for longer arcs. Multiple shooting splits the orbit into N independently integrated segments and imposes continuity constraints at the patch points:

$$\mathbf{x}_i(t_{i+1}) = \mathbf{x}_{i+1}(t_{i+1}), \quad (4)$$

adjusting all segment initial states (optionally including time nodes) simultaneously—turning “long-arc sensitivity” into “many small corrections”. The standard implementation reaches a tolerance of 10^{-10} .

Patch-point correction under the ephemeris model uniformly goes through the Rust multiple-shooting kernel, dispatched by orbit stability along two paths:

- **Stable orbits** (DRO): a velocity-weighted two-level path—after correcting one revolution, free extrapolation remains bounded;
- **Unstable orbits** (Halo/NRHO/DPO): segmented full-arc shooting, integrating segment by segment to fill the ephemeris grid; NRHO fixes 1 revolution per segment, Halo at most 3, and DPO uses 64 equally-spaced patch points per revolution.

Node sampling is likewise tailored to orbital physics: Halo uses perilune-clustered sampling (5 nodes inside the perilune window plus 8 outside), because the STM is ill-conditioned where the velocity is large near perilune; NRHO uses equal-time sampling.

4.5 Continuation: From One Orbit to a Family

Given a converged seed orbit, a family is generated by stepping along a family parameter (Jacobi constant, amplitude, etc.), reconverging with differential correction at each step. Two continuation schemes are provided:

- **Natural-parameter continuation**: advances along a single parameter, each step seeded by the current solution;
- **Pseudo-arclength continuation (PAL)**: adds an arclength constraint and can pass turning points, suitable for families with non-monotonic parameter–amplitude relations (e.g. the Halo amplitude–period curve).

Failure handling is engineered: a non-converged step retries with a reduced step size; if convergence still fails at the lower bound, that direction terminates; only converged members are appended to the family, with diagnostics reporting total/successful/failed steps. The L2 NRHO family uses a dedicated “fold then continue with fixed x_0 ” arrangement; the Halo family uses natural continuation with fixed z_0 , positive and negative z_0 giving the northern and southern families.

4.6 Stability Analysis and Invariant Manifolds

The stability of a periodic orbit is determined by its monodromy matrix (the STM over one period):

$$\mathbf{M} = \Phi(T, 0). \quad (5)$$

By the symplectic structure of Hamiltonian systems, eigenvalues appear in reciprocal pairs $(\lambda, 1/\lambda)$. The Broucke stability index sums each reciprocal pair:

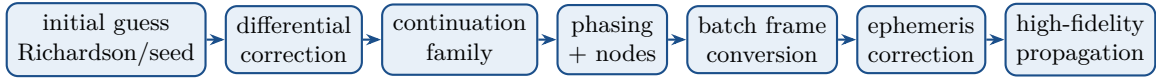
$$\nu = \lambda + \frac{1}{\lambda}, \quad (6)$$

with $|\nu| < 2$ stable and $|\nu| > 2$ unstable (the larger, the more unstable). The relation of eigenvalues to $+1$, -1 , and the unit circle identifies bifurcation types (saddle-node / period-doubling / torus), with family-level bifurcation detection supported. Numerical trustworthiness is self-checked via symplectic residuals: if $|\det \mathbf{M} - 1|$ or $\|\mathbf{M}^\top \mathbf{J} \mathbf{M} - \mathbf{J}\|$ is significantly large, the integration tolerance should be tightened and the analysis redone.

The real eigenvectors of the monodromy matrix also give the invariant manifold directions: the stable manifold takes the $|\lambda| < 1$ direction and the unstable manifold the $|\lambda| > 1$ direction. Transporting the eigenvectors along the orbit by the STM and applying a $\pm\epsilon$ perturbation (typically 50 km/DU) yields manifold seeds; integrating backward/forward produces the manifold tubes used for low-energy transfer stitching (Section 5).

4.7 The Design Pipeline

Chaining these algorithms together, the complete pipeline of one mission orbit is:



All numerical heavy lifting (propagation, shooting, batch frame conversion, batch ephemeris queries) runs on the Rust side, with Python only orchestrating; the end-to-end generation of four typical orbits (LLO, ELFO, DRO, Halo) takes under one minute in total with a release build.

NominalOrbit: The Contract Between Design and Control

A designed nominal orbit is delivered as a `NominalOrbit` structure: an equally-spaced state table + a Floquet basis + projection factors. Station-keeping algorithms consume this structure directly—the Floquet analysis produced during design does not need to be recomputed during control; the two stages are decoupled through a single data contract.

5 Transfer Orbit Design

Transfer design in e2m2e covers impulsive, low-energy, and low-thrust propulsion modes. The core methodology is a two-step “search then optimize” scheme: ballistic propagation on a coarse grid first filters the feasible solution families, then the best candidates are handed to nonlinear programming (NLP) for refinement. The unified design variables are the tangential velocity ratio α (controlling the departure direction), the transfer time T , and the time t_{ins} from apoapsis to the insertion point, with the total objective

$$J = \Delta v_1 + \Delta v_2, \quad (7)$$

subject to position continuity at the insertion point, velocity-direction parallelism with the target, and collision avoidance with the Earth and the Moon. The dynamical baseline is the CR3BP, with μ taken from the DE421 constant baseline.

5.1 Impulsive Transfers

5.1.1 Lambert Solving and Porkchop Scans

The Lambert solver implements the Izzo (2015) algorithm with a Rust kernel (Python only converts and wraps): it supports short-/long-way directions and multi-revolution solutions (returning the low-energy right branch), and provides a batch interface—a grid of N geometries $\times M$ times of flight enters Rust in one call, with infeasible combinations filled with NaN without affecting the rest. The solver is verified against Vallado’s classic test case.

Solving Lambert point by point over a departure-epoch $\times$ time-of-flight grid yields the porkchop two-impulse ΔV map; terminal states are extracted through the terminal-condition abstraction (`TerminalCondition`), and analytic terminals need no dynamics propagation.

5.1.2 Three-Body Targeting

A two-body Lambert solution only qualifies as an initial guess in cislunar space. The three-body Lambert targeter seeds a damped Newton iteration in the CR3BP with the two-body solution: propagating with the state transition matrix $\Phi(t, t_0)$, each correction is solved from the Φ_{rv} block of the terminal STM; the step is halved whenever a full step increases the error; convergence is declared when the terminal position error falls below 10^{-8} (dimensionless).

5.1.3 Multi-Impulse Optimization and the Lawden Primer-Vector Test

For an n -impulse transfer between fixed endpoints, the decision variables are only the intermediate nodes $[t_i, \mathbf{r}_i]$; arcs between adjacent nodes are closed by Lambert solutions (optionally refined by three-body targeting), and SLSQP minimizes the total ΔV . Whether a two-impulse solution can still be improved is answered by the Lawden primer-vector test: from the transversality conditions $\mathbf{p}(t_0) = \hat{\Delta}v_0$ and $\mathbf{p}(t_f) = \hat{\Delta}v_f$, the costate is carried by the STM to give the $\mathbf{p}(t)$ curve, with the necessary optimality condition

$$\|\mathbf{p}(t)\| \leq 1 \quad \text{throughout}, \quad \|\mathbf{p}\| = 1 \text{ at impulses, aligned with them.} \quad (8)$$

If $\|\mathbf{p}\| > 1$ within an arc, inserting an intermediate impulse reduces the total ΔV (Lion & Handelsman, 1968); the test also reports the suggested insertion time and position, directly seeding a three-impulse optimization. A LEO→GEO example reproduces the Hohmann benchmark of 3.7708 km/s.

The classical Hohmann two-impulse minimum-energy transfer

$$\Delta v_1 = \sqrt{\frac{\mu}{r_1}} \left(\sqrt{\frac{2r_2}{r_1 + r_2}} - 1 \right), \quad \Delta v_2 = \sqrt{\frac{\mu}{r_2}} \left(1 - \sqrt{\frac{2r_1}{r_1 + r_2}} \right) \quad (9)$$

is provided as an analytic benchmark and a source of initial guesses.

5.2 Low-Energy Transfers

5.2.1 Lunar Gravity Assist (LGA)

A three-leg conic-patching design: hyperbolic departure from the initial orbit $\rightarrow$ a gravity assist inside the lunar sphere of influence that rotates the velocity vector $\rightarrow$ hyperbolic insertion into the target orbit. The search interface works in dimensionless CR3BP states; a typical search domain is 15–45 days of flight time with a total Δv cap of 4.0 km/s, and results report total cost, flight time, and perilune altitude.

5.2.2 WSB Ballistic Capture

The weak stability boundary (WSB) route exploits the Sun–Earth weakly stable region: the spacecraft leaves the lunar sphere of influence onto a distant Sun-perturbed arc, where solar gravity does work near apoapsis so that the Moon-relative velocity drops below the capture threshold, yielding ballistic capture at near-zero Δv (Belbruno & Miller, 1993). The capture criterion is a negative two-body Kepler energy at perilune:

$$H_2 = \frac{\|\mathbf{v}\|^2}{2} - \frac{\mu_{\text{moon}}}{r} < 0. \quad (10)$$

The solution proceeds in two steps: a **ballistic grid search** propagates forward in the BCR4BP over a Sun-phase $\times$ departure-phase $\times$ flight-time grid, filtering by perilune altitude and the H_2 criterion—propagation, section detection, and filtering all run in Rust with Rayon parallelism (the Python reference implementation runs only when explicitly requested, never as an automatic fallback); an **arrival-leg refinement** then closes the best candidate’s Moon-centered arrival arc with three-body Lambert targeting. A typical search domain is 90–150 days.

5.2.3 Invariant-Manifold Stitching

A third low-energy route exploits the invariant manifold tubes of periodic orbits themselves: crossings of the departure orbit’s unstable manifold and the target orbit’s stable manifold on the same Poincaré section (a planar section, or a periapsis section $s = \mathbf{r} \cdot \mathbf{v} = 0$) are paired and ranked by the weighted patching cost

$$w_r \|\Delta \mathbf{r}\| + w_v \|\Delta \mathbf{v}\|. \quad (11)$$

The end-to-end pipeline: propagate the departure unstable manifold $\times$ target stable manifold (all four $\pm$ branch combinations, globally best taken) to the secondary’s periapsis section, pick the best patch, then close onto the target orbit after the patch point with three-body Lambert targeting; impulses accordingly split into departure, patch, and arrival segments. In the benchmark case (medium-to-large-amplitude members of the L1 Lyapunov family), the patch impulse is on the order of tens of m/s. The route is currently CR3BP-based, with an ephemeris interface reserved.

The value of low-energy routes: trading time for propellant, enabling economical lander/orbiter scenarios and even emergency low-fuel rescue corridors.

5.3 Low-Thrust Transfers

The low-thrust leg uses the 7-dimensional augmented state $[\mathbf{r}, \mathbf{v}, m]$:

$$\dot{\mathbf{r}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \mathbf{a}_{\text{grav}} + \frac{T \delta}{m} \hat{\mathbf{u}}, \quad \dot{m} = -\frac{T \delta}{I_{sp} g_0}, \quad (12)$$

with throttle $\delta \in [0, 1]$ and thrust direction $\hat{\mathbf{u}}$ parameterized by spherical angles (θ_1, θ_2) . Solving is a two-level loop: a **Q-law initial guess** integrates a Lyapunov feedback law forward to produce a suboptimal control history; an **SLSQP polish** then minimizes propellant over piecewise-constant controls $(\delta, \theta_1, \theta_2)$ per segment, matching the terminal position and velocity. Two solvers are available: shooting (analytic Jacobians, 5–24 $\times$ faster) and Hermite–Simpson collocation (more robust at large scale).

Impulsive and low-thrust costs are distinguished via the equivalent Δv : impulsive uses $\Delta v = \sum_i \|\Delta \mathbf{v}_i\|$, while low-thrust inverts the Tsiolkovsky equation, $\Delta v_{\text{eq}} = I_{sp} g_0 \ln(m_0/m_f)$; due to gravity losses, $\Delta v_{\text{eq}} \geq$ the impulsive ΔV , with significant differences on high-energy transfers.

5.4 The Grid-Search + NLP Two-Step Optimization Framework

Step one, **grid search**: departure points are sampled equally in time along the departure orbit (typically 200), each with α uniformly sampled in $[0.5, 2.5]$ (typically 101); transfer arcs are integrated forward and filtered by collision (200 km at Earth / 100 km at the Moon), intersection, and minimum-distance (10^{-3}) constraints; the numerical kernel has sunk to Rust and runs in parallel.

Step two, **NLP refinement**: decision variables $\mathbf{y} = \{\alpha, T, t_{\text{ins}}\}$, objective $J(\mathbf{y}) = \Delta v_1 + \Delta v_2$, constraints of insertion-point position continuity (equality) and velocity parallelism (equality, or relaxed to an inequality with an angular tolerance of 0.05 rad). The solver defaults to SciPy SLSQP, with the COPT commercial optimizer as an option and fallback on failure.

Table 5: Transfer design capabilities at a glance

Category	Method	Notes
Impulsive	Lambert (Izzo)	Rust kernel, batch grid solving, short/long way and multi-revolution
Impulsive	Three-body targeting	Damped Newton + STM, position error $< 10^{-8}$ at convergence
Impulsive	Multi-impulse	SLSQP + Lawden primer-vector test, automatic impulse-insertion suggestions
Impulsive	Hohmann	Analytic benchmark and initial guesses
Low-energy	LGA	Three-leg conic patching, TOF 15–45 days
Low-energy	WSB	BCR4BP + $H_2 < 0$ criterion, Rust Rayon parallel search
Low-energy	Manifold stitching	Poincaré section pairing + three-body closure, patch impulses at m/s scale
Low-thrust	Q-law + NLP	Shooting (analytic Jacobian, 5–24 $\times$) or collocation
Framework	Search-optimize	Grid search (Rust parallel) $\rightarrow$ SLSQP/COPT refinement

6 Orbit Control and Monte Carlo Simulation

Unstable orbits (Halo/NRHO/DPO) are only truly usable together with station keeping. e2m2e makes control simulation the last link of the design chain: the NominalOrbit produced during design (an equally-spaced state table + Floquet basis + projection factors) is consumed directly by the control algorithms, with no need to recompute the Floquet analysis.

6.1 Three Control Laws

- **Feature-point control**: corrections are applied only at characteristic locations (e.g. perilune or plane crossings)—few maneuvers, low propellant cost, suitable for long-term keeping;
- **Strict target-point control**: the state is driven exactly back to a target point on the nominal orbit—highest tracking accuracy;
- **Relaxed target-point control**: the target point admits an error band, trading tracking accuracy against propellant consumption.

Angular-momentum management is also provided: joint attitude/thruster control that folds momentum unloading into the station-keeping strategy.

6.2 Monte Carlo Error Simulation

Station-keeping propellant statistics must be evaluated under error models. e2m2e supports Monte Carlo simulation with jointly injected navigation and thrust errors: batches of keep-

ing simulations on the same nominal orbit yield ΔV consumption distributions and orbit-maintenance statistics, giving mission studies confidence intervals rather than point estimates. Simulation artifacts are automatically archived in the orbit catalog with tag management—in the MCP scenario, “run 100 Monte Carlo station-keeping simulations on this NRHO and tag the results candidate” is a workflow completed by a single natural-language instruction.

7 Interfaces, Data, and Ecosystem

7.1 The Facade: A Single Task-Level Entry

All capabilities converge on a single **Facade** class (ADR 0014), from which the CLI and the MCP server are derived from the same metadata source: beyond input validation, the task-level API model also exposes parameter metadata for the current task context (conditional value domains and validators share one rule definition), so that GUIs, the CLI, and MCP can generate widgets and range hints—external consumers keep no local copies of parameter rules. Task-level methods come in three tiers:

- **Primary tasks:** `design_orbit` (mission orbit design: CR3BP initial guess → ephemeris correction → high-fidelity propagation), `control_orbit` (station-keeping Monte Carlo simulation), `transfer_design` (HMN/LGA/WSB/low-thrust), `orbit_propagation` (orbit propagation), and `spacetime_transform` (synodic↔J2000 and GCRS↔EBCRS conversions);
- **Secondary subtasks:** `orbit_family_generation` (continuation generation of eight families, with family records automatically archived), among others;
- **Orbit catalog:** 7 catalog methods (below).

Listing 1: Designing an Earth–Moon L2 Halo orbit (examples/main_design.py)

```

1  from e2m2e.api import Facade
2
3  facade = Facade()
4
5  result = facade.design_orbit(
6      orbit_type="Halo",
7      collinear_point=2,
8      amplitude=30000.0,          # km
9      epoch=[2024, 1, 1, 0, 0, 0.0],
10     duration=365.25 * 86400.0,  # s
11 )
12
13 print(result.orbit_type)
14 print(result.initial_state)

```

7.2 MCP: An Agent-Native Interface

e2m2e can act as an MCP server (`e2m2e mcp-serve`, stdio transport, listening on no port), exposing 13 task-level tools to LLM agents: 6 task methods (the 5 primary ones plus family generation) and 7 orbit-catalog tools. The tool list is purely derived from Facade method metadata—nothing else to maintain. All responses use a unified envelope `{status, data, error, meta}`: errors are expressed as error codes (`INVALID_PARAMS`, `RECORD_NOT_FOUND`, `PROPAGATION_FAILED`, etc.) without leaking tracebacks, and numerical tools report soft-failure semantics via `data.status` (converged / diverged / stagnated / max_iterations, ADR 0020)—an agent can treat “not converged” as normal information rather than an exception.

Setup is a single command registered in the MCP client; it is recommended to pin `cwd` to the directory containing `kernels/` (SPICE kernels) and `catalog/` (the orbit catalog), or to give absolute paths via `SPICE_KERNEL_DIR / E2M2E_CATALOG_DIR`. Once configured, the server can be driven in natural language, e.g.:

Design an L2 southern NRHO with 3000 km perilune altitude, then run 100 Monte Carlo station-keeping simulations on it and tag the results “candidate”.

7.3 The Orbit Catalog

Computation artifacts are automatically archived (ADR 0031): a record is JSON metadata + NPZ array segments (with a schema version), with SQLite only as a derived index. Productive tools return a `record_id` on success, which chains across tools (e.g. `control_orbit` can reference the output of `design_orbit` via `input_record_id`). Seven catalog tools cover the full lifecycle: `catalog_query` (multi-dimensional filtering by family / libration point / Jacobi range / amplitude / tag / convergence status), `catalog_get`, `catalog_delete`, `catalog_tag` (teaching annotations), `catalog_promote` (promoting a family member to a standalone record), `catalog_export` (packaged export), and `catalog_sweep` (batch generation over parameter grids).

7.4 Installation and Deployment

Listing 2: Installation and SPICE kernel setup

```

1 uv pip install e2m2e                # release wheels embed CSPICE, ready
   out of the box
2 uv pip install "e2m2e[mcp]"         # MCP server
3
4 # development from source (requires the Rust toolchain):
5 git clone https://github.com/cislunarspace/e2m2e.git
6 cd e2m2e
7 make dev                          # sync deps + fetch CSPICE package & SPICE kernels +
   build the Rust extension

```

Release wheels cover Windows x86_64, Linux x86_64, and Linux aarch64 (Kunpeng / Phytium / Raspberry Pi), with statically linked CSPICE embedded, so the Rust fast paths (STM propagation, shooting, third-body gravity, etc.) work out of the box; when the extension is unavailable, an explicit error is raised rather than silently degrading. The SPICE kernels are packaged as the GitHub Release asset `kernels-v1` (downloaded automatically by `make setup`), including DE planetary ephemerides, Earth rotation, Moon attitude, leap seconds, and planetary constants kernels; the r2s2 spacetime transformation additionally requires an ephemeris containing the TT–TDB time ephemeris. The official source is NASA NAIF.

The `examples/` directory provides six runnable examples: end-to-end orbit design, station-keeping Monte Carlo, high-fidelity propagation, a Lambert transfer, a four-layer comparison of lunar orbits (LLO/ELFO/DRO/Halo, under one minute in total), and orbit-family generation.

8 Comparison with Existing Tools

8.1 Scope and Method

This chapter positions e2m2e within the existing aerospace software ecosystem. Ten comparison subjects cover four shapes: general-purpose mission analysis platforms (GMAT, STK, ATK),

open-source astrodynamics libraries (Orekit, pykep, poliaastro), the starting point of the self-built route and its accuracy yardstick (SciPy), and ecosystem complements (r2s2, Cesium, ootk). The survey was completed in August 2026 with e2m2e 5.8.8 as the version anchor: GMAT, Orekit, pykep, poliaastro, ootk, and r2s2 were examined at the code level (repository-wide greps and source reading), while the closed-source STK and ATK and the pure-rendering Cesium were surveyed through official documentation and third-party literature. Every number cited below can be re-verified in the respective repositories or official documentation.

Three principles keep the narrative fair:

1. **No head-to-head, no multiples:** where same-scenario benchmark data are lacking (STK, Orekit, ATK), only architectural comparisons are made—no performance multiples are claimed;
2. **Partitioned performance claims:** speedups against the SciPy self-built route can be quantified (the callback overhead is structural); differences from GMAT are stated at the parallelism architecture level;
3. **Accuracy as alignment, not contest:** e2m2e’s force-model layer uses GMAT’s mathematical specification as its porting baseline, so the accuracy relationship is “alignment statement + component porting consistency”, not “who is more accurate”.

8.2 Ecosystem Landscape

Table 6: The comparison landscape (version anchors, August 2026)

Tool	Vendor / License	Positioning (language)	Relationship to e2m2e
GMAT	NASA / Apache-2.0	General mission analysis platform, GUI + plugins (C++, ~850k LoC)	e2m2e’s force-model alignment baseline; no CR3BP kernel
STK	Ansys / proprietary	Digital mission engineering platform	Industry de facto standard; Astrogator has a CR3BP framework but no family automation
ATK	NUDT / closed binary	Domestic STK counterpart, 300+ organizations	General MCS framework; no declared three-body design tools
Orekit	CS GROUP / Apache-2.0	Flight dynamics infrastructure library (Java, ~470k LoC)	The most mature infrastructure; three-body limited to Halo/Lyapunov components
pykep	ESA / MPL-2.0	Interplanetary trajectory optimization research library (C++23 + pybind)	Three-body limited to integrators and benchmark problems; not operational
poliaastro	community / MIT	Teaching-grade astrodynamics (Python + numba)	Archived (Oct 2023); three-body stuck in contrib drafts
SciPy	community / BSD family	General scientific computing (Python + C/Fortran)	Starting point of the self-built route and e2m2e’s accuracy yardstick
ootk	thkruz / AGPL-3.0	Web Earth-orbit SSA toolkit (TypeScript)	Different track; near-zero overlap; complementary
r2s2	PMO CAS / GPL-3.0	Relativistic spacetime coordinate conversion (Python, ~1.2k LoC)	e2m2e’s underlying dependency , not a competitor
Cesium	Cesium GS / Apache-2.0	3D visualization engine (JS/TS, WebGL)	Pure rendering, zero computation; a natural front end

8.3 The Core Differentiation Axis: Cislunar Three-Body Design Automation

The watershed between e2m2e and every comparison subject is whether the three-body capability is a *design pipeline* or merely *parts and frameworks* (Table 7).

Table 7: Cislunar three-body capability comparison (the core differentiation axis)

Tool	CR3BP dynamics / family generation	Low-energy transfer / station keeping / ephemeris correction
e2m2e	Native three-model (CR3BP/BCR4BP/ephemeris N -body) conversions; automatic generation of eight families + continuation + energy-window batch scanning	Manifold stitching + LGA/WSB; three control laws + Monte Carlo; the initial-guess $\rightarrow$ ephemeris correction (20 m convergence) $\rightarrow$ high-fidelity three-stage chain
GMAT	No CR3BP module (repository-wide grep: zero hits); libration points are geometric points only, rotating frames assembled by users	None; the ephemeris propagator is a replay-style propagator
STK	Astrogator has a built-in CR3BP force model + differential targeting since 11.7	No templated family generation (no public evidence found); low-energy transfers assembled in MCS; no keeping control laws
ATK	High-precision N -body numerical propagation + general MCS (11 segment types $\times$ 60 constraints)	Family generation/manifolds not claimed in public materials; a paper reproducing GMAT’s Halo keeping case
Orekit	CR3BPForceModel + STM, component-style	Only Halo/Lyapunov two types; manifolds as single-point initial values only; no keeping; shooter accepts ephemeris models (user-assembled)
pykep	heyoka symbolic factory + 42-dimensional variational equations	No family generation (grep: zero hits); no manifolds; 14 cislunar low-thrust benchmark problems —whose inputs are precisely e2m2e’s outputs
Others	SciPy: zero orbit knowledge; poliastro: three-body in contrib drafts; ootk/Cesium: no three-body code	—

The conclusion is clear: **none of the eight comparison subjects combines “eight-family generation + manifold low-energy transfer + three-body station keeping + ephemeris correction chain”**. GMAT’s three-body capability is geometric points plus N -body summation; STK provides a framework whose workflows are hand-built sequences; Orekit provides parts that users must assemble; pykep treats the three-body problem as an optimization carrier rather than a design object; poliastro’s three-body work never entered the main library. Cislunar three-body design automation is therefore e2m2e’s unique niche.

8.4 Profiles of Representative Subjects

8.4.1 General-Purpose Platforms: GMAT, STK, ATK

GMAT (NASA, ~850k lines of C++) is the “closest” subject to e2m2e: the PD78/RK89 weights, the conical shadow model, the three relativistic terms, the STM step-error-control semantics, and the EOP/leap second fixtures are all ported from the GMAT R2026a mathematical specification—the force-model layer can be described as “a Rust re-implementation and extension of GMAT’s mathematical specification”. In return, GMAT’s solver and optimization ecosystem (DifferentialCorrector, Yukon SQP, the CSALT direct-collocation framework, Bulirsch–Stoer integrators) and its document-level V&V system (mathematical specification, flight-qualification documents) are thickness e2m2e does not yet have; its lack of a parallel propagation kernel (zero grep hits in physics loops) marks a generational difference from e2m2e’s service-oriented form. The two are in a “benchmark and differentiate” relationship: GMAT for cross-checking and operations analysis, e2m2e for the three-body design pipeline.

STK (Ansys, proprietary) is the industry’s de facto standard. Astrogator has included the CR3BP force model and differential targeting since 11.7, and the Artemis II Orion tracking visualization is STK-driven. The comparison reduces to four axes: license (proprietary commercial vs Apache-2.0), three-body automation (framework vs one-click family generation), cost (a government purchase record of USD 114.5k for a single Pro token vs free), and programmability (an Integration license module vs a native MCP server). STK’s coverage/link/scheduling/ODTK module ecosystem is outside e2m2e’s scope; in typical workflows the two form an “open-source computation + commercial presentation” layering.

ATK (NUDT, closed binary) takes the “Chinese counterpart of STK” route: a building-block maneuver planning framework (11 orbit-segment types, 19 stopping conditions, 11 categories / 60 constraint components, unified NLP with three solver classes—differential correction / SQP / intelligent optimization), backed by operational support of the Tiangong station, Shenzhou and Tianzhou missions. Its paper reproduces the lunar free-return case and GMAT’s built-in L2 Halo keeping case (four maneuvers of $-0.25/1.64/8.05/6.44$ m/s), showing the framework can carry three-body problems; yet public materials declare no CR3BP propagator, no libration-point family generation, and no invariant-manifold automation—which remains e2m2e’s cleanest point of difference. “Platform vs library”, “closed vs open”, and “general MCS vs three-body-dedicated pipeline” are the three axes of comparison.

8.4.2 Open-Source Astrodynamics Libraries: Orekit, pykep, poliastro

Orekit (CS GROUP, ~470k lines of Java, 24 years of development since 2002) is the benchmark for engineering-grade infrastructure: frame/time chains compared element-by-element against the official SOFA library to 10^{-12} , a propagator family spanning numerical / analytic / DSST semi-analytic / SGP4 / GNSS, a full orbit-determination suite (batch least squares + EKF/UKF), and 80+ event detectors—depth e2m2e does not have. But its three-body capability is component-style: the dynamics model, Richardson initial guess, and shooter parts exist, while the orbit-type enumeration holds only Halo and Lyapunov—no family continuation, no

Lissajous/DRO, no batch manifold propagation and stitching, no keeping control laws. Both Apache-2.0, “Orekit for the Earth segment and orbit determination, e2m2e for cislunar design” is a natural division of labor.

pykep (ESA, `kep3`, MPL-2.0) is an interplanetary trajectory optimization research library: the `heyoka` LLVM-JIT Taylor integrator (tolerances to 10^{-16}) plus `pagmo` evolutionary optimization, MGA encodings, Sims–Flanagan/ZOH low-thrust transcriptions, and Pontryagin indirect methods—officially “not an operational flight dynamics tool”. Its three-body capability stops at a symbolic dynamics factory and 14 cislunar low-thrust benchmark problems (TOPS: Halo-to-Halo, DRO-to-NRHO, etc.); libration-point computation, family construction, monodromy matrices, and manifold targeting are unimplemented—while the inputs of the TOPS benchmarks (Halo/DRO/NRHO nominal orbits) are precisely e2m2e’s outputs. “e2m2e generates the nominal orbit $\rightarrow$ pykep + `pygmo` for low-thrust global optimization $\rightarrow$ e2m2e for high-fidelity verification and keeping” is a ready-made division-of-labor chain.

poliastro (MIT, archived October 2023) matters to this white paper as **evidence of an ecosystem vacancy**: its unit-safe API and plotly visualization are teaching-grade gems, but its cislunar three-body work remained in the contrib draft directory (its own README lists as unimplemented: differential correction, family continuation, the third-order Halo analytic solution, monodromy, manifolds), with only 6 perturbation functions and test tolerances at the $10^{-4} \sim 10^{-2}$ level. The most active teaching-oriented astrodynamics library in the Python ecosystem is archived with its three-body agenda unfinished—precisely the vacancy e2m2e fills.

8.4.3 The Self-Built Route and Accuracy Yardstick: SciPy

SciPy knows nothing about orbits—no time systems, no reference frames, no ephemerides, no force models—yet `solve_ivp` + `root` assembled into a CR3BP targeter is the de facto self-built route in academia, and the “lower-bound alternative” an e2m2e user would try first. On performance, its explicit RK family is implemented in Python with a right-hand-side callback at every step, running 1–3 orders of magnitude slower than compiled kernels on tight-tolerance, batch-targeting workloads (community consensus; e2m2e’s corresponding evidence: the same parallelized problem yields about 1/10 of the Rust-sunk speedup under Python multiprocessing). On accuracy the two are at par—e2m2e’s test suite uses `scipy` DOP853 (10^{-13}) as its gold-standard reference, with a LEO+J2 one-day-arc consistency of $< 10^{-9}$. The gap is not accuracy but the fact that reaching the same accuracy requires users to build the entire domain layer themselves.

8.4.4 Ecosystem Complements: `r2s2`, Cesium, `ootk`

r2s2 (Purple Mountain Observatory, CAS; ~1150 lines of pure Python) is not a competitor but an underlying dependency: the GCRS $\leftrightarrow$ EBCRS relativistic spacetime conversion rides on it, with reverse conversions converging to 1 ns / 1 mm by default. e2m2e’s independent verification of it (round-trip consistency, differencing against the SPICE chain, differencing against ERFA `dtddb`) constitutes its main test coverage. `r2s2` is GPL-3.0; closed-source integrators should note the copyleft implications.

Cesium (CesiumJS, Apache-2.0) is a pure rendering layer with zero orbital mechanics—satellite positions must be fed in externally. It was historically “built to track objects in space” and treats cislunar-scale scenes as first-class citizens (ICRF inertial rendering; the Artemis II lunar-orbit tracking case). Visualization is exactly e2m2e’s weak spot, and “EphemerisTable $\rightarrow$ CZML $\rightarrow$ Cesium” is a license-clean composition path (Orekit + OreCZML already validate the same template). The division to remember: Cesium faithfully renders whatever trajectory it is fed; the correctness of the trajectory is e2m2e’s responsibility.

ootk (TypeScript, AGPL-3.0) plays a different sport entirely: SGP4/TLE cataloging, sensors, conjunctions, Web-side SSA—near-zero overlap with e2m2e. Its value is twofold: complementarity (catalog/observation ground to ootk, high-fidelity mission design to e2m2e), and methodology: ootk regresses against bit-level snapshots of the official AFSPC SGP4Prop and Vallado’s verification vectors, with SLR precise ephemerides as independent ground truth—the kind of “bit-level alignment against authoritative reference implementations” that e2m2e’s verification system can borrow as it matures.

8.5 Performance and Accuracy: Partitioned Evidence

Table 8: Parallelism architecture and public quantitative figures

Tool	Architecture	Public figures
e2m2e	Rust kernel + Rayon (shooting/grid/manifolds/WSB)	Reproducible; benchmark scripts in-repo
GMAT	No parallel propagation kernel (single-instance sequential)	none
STK	Parallel Computing multi-core distribution	8 cores, 180 samples, 70 s (only figure found)
Orekit	PropagatorsParallelizer	no public figures
pykep	heyoka LLVM JIT + per-tolerance caching	SF evaluation 5.25 μ s; “thousands of trajectories per second”
SciPy	single-threaded, Python RHS callback	1–3 orders slower than compiled kernels
ootk	pure JS, single-threaded (light compute $\times$ massive calls)	SGP4 391 ns/call

e2m2e’s own performance figures are all reproducible: transfer grid search accelerated 5.5–10.4 $\times$ by sinking to Rust; 180-day Halo end-to-end design 223 s $\rightarrow$ 63 s; shooting 9.4 $\times$ faster with SRP on the Rust STM fast path; low-thrust analytic Jacobians 3–24 $\times$. Per the principles above, these numbers are compared in magnitude only against the SciPy self-built route and at architecture level against GMAT; no mutual multiples are claimed against STK/Orekit/ATK—none of them publishes same-scenario comparison data.

On accuracy (Table 9), anchors and verification philosophies differ: Orekit’s and ootk’s element-wise/bit-level comparisons against authoritative references have the most complete evidence chains; pykep reaches 10^{-13} in dimensionless normalized units; e2m2e achieves $10^{-9} \sim 10^{-11}$ for integrators against analytic solutions, $< 10^{-12}$ for the coordinate chain, and sub-100 m force-model alignment with DFH (88 m over a 7-day Halo arc)—though, per the verification-by-definition principle of ADR 0013, the external comparison tests have been removed from the repository, leaving only self-runnable GMAT comparison scripts. Accuracy is not the differentiator; the solvable problem domain is.

8.6 The Composition Ecosystem: A Cislunar Design Engine

Shifting the viewpoint from competition to composition, e2m2e sits in the ecosystem map as a **cislunar design engine**: upstream it is fed by spacetime and ephemeris suppliers (NASA SPICE/DE ephemerides, the r2s2 relativistic spacetime conversion), and downstream it has ready-made open-source consumers (Figure 2)—Cesium for rendering, GMAT for cross-checking and operations analysis, Orekit for the Earth segment and orbit determination, and pykep for low-thrust global optimization (whose cislunar benchmark problems take e2m2e’s

Table 9: Accuracy verification anchors

Tool	External ground-truth anchor	Figures
e2m2e	DFH / GMAT fixtures / scipy DOP853 / analytic solutions	Halo 7-day 88 m; integrators $10^{-9} \sim 10^{-11}$; ITRF $< 10^{-12}$
GMAT	V&V documentation + institutional endorsement	default per-step tolerance 10^{-11}
STK	third-party papers / community cross-checks	RKF78 per-step tolerance configurable to 10^{-13}
ATK	paper cases (incl. GMAT Halo case reproduction)	100 m targeting achieved to 100.23 m
Orekit	element-wise comparison vs official SOFA library	frame conversion 10^{-12}
pykep	hardcoded high-precision truth values	CR3BP $\leq 10^{-13}$ (tol. 10^{-16})
ootk	AFSPC bit-level snapshots + Vallado vectors + SLR ephemerides	SGP4 5×10^{-9} km
r2s2	(tested by its dependent e2m2e)	iterative convergence 1 ns / 1 mm

outputs as inputs). Licenses coexist along every chain (with the GPL-3.0 note for r2s2 in closed-source integration).

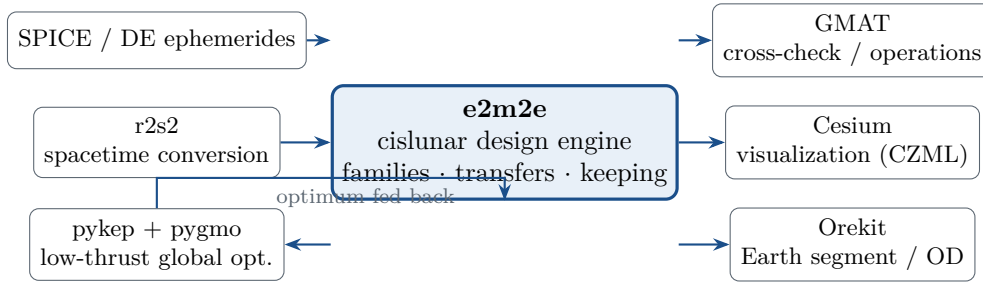


Figure 2: The composition ecosystem of e2m2e: spacetime/ephemeris suppliers upstream; rendering, cross-checking, orbit determination, and global optimization consumers downstream—e2m2e holds exactly the cislunar design engine layer.

Summary: A Repeatedly Verified Vacancy

The general platforms have no three-body kernel (GMAT), or a framework without automation (STK/Astrogator), or components without a pipeline (Orekit), or dynamics without design tools (pykep), while the teaching library was archived with its three-body work still in contrib drafts (poliastro)—none of the eight subjects combines “eight-family generation + manifold low-energy transfer + three-body keeping + ephemeris correction chain”. This vacancy, verified subject-by-subject at the code level, is e2m2e’s niche.

9 Verification and Conclusion

9.1 Verification by Definition

e2m2e’s verification philosophy is *verification by definition* (ADR 0013): assertions land mainly on analytic solutions, conserved quantities, and literature formulas, rather than on regression comparisons “consistent with its own previous output”. The test suite is divided into six

categories by verification target, with directories mirroring the source layout:

- **algorithm**: orchestration chains of coordinates, dynamics, design, correction, and transfer;
- **api**: Facade and MCP interface responses and error translation;
- **data**: the data layer—kernels, reference frames, physical constants;
- **mbse**: MBSE data models, requirement registration, and diagram generation;
- **numerical**: integrator accuracy against analytic orbits; force-model accelerations and Jacobians;
- **tools**: auxiliary utilities—logging, formatting, visualization.

The coordinate and time chains are additionally cross-compared against GMAT reference data; Rust-side numerical methods are verified against analytic solutions in `crates/*/tests/`. High-fidelity propagation accuracy aligns with GMAT and DFH to the sub-100 m level. The Lambert solver is verified against Vallado’s classic test case; multi-impulse optimization reproduces the LEO→GEO Hohmann benchmark of 3.7708 km/s; differential correction, continuation, and stability analysis are checked against formulas from Broucke, Richardson, Gómez, and others.

9.2 Conclusion and Outlook

e2m2e converges the complete chain of cislunar mission orbit design—the spacetime reference, multi-fidelity dynamics, periodic orbit family generation, three classes of transfer design, and station-keeping simulation—under a single task-level Facade, guarantees the performance and parallel safety of heavy iteration with a Rust numerical kernel, and opens the entire chain to LLM agents through the MCP interface. It does not pursue the full coverage of general-purpose tools; instead, in the sufficiently deep domain of cislunar space, it makes every step from initial guess to verification a composable, traceable, and auditable engineering component.

On the roadmap: the unified ephemeris-serialization intermediate format `EphemerisTable` with converters to/from CCSDS OEM/ODM, SPICE BSP/PCK, and other typical formats; further migration of remaining Python numerics (such as NLP optimization) into the Rust kernel; and, building on the orbit catalog as a data foundation, extensions toward intelligent game-theoretic research.

Repository	https://github.com/cislunarspace/e2m2e
Documentation	https://cislunarspace.github.io/e2m2e/
License	Apache 2.0
Install	<code>uv pip install e2m2e</code>